WORKED SOLUTION

STEP I 2019, Question 11

Probability and Statistics • Geometric Distribution

STEP I 2019, Question 11

Read the question before you go on. It is in the STEP I paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 12.9, *A Fair Decision from a Biased Coin*: https://beyondstep.keithmcnulty.org/ch12/von-neumann.html

The key idea

Each round of the two-person procedure is an independent trial that either ends things or doesn't, so the round of the decision follows a geometric distribution. The bound "whatever the value of p" then comes down to how large pq can be when $p+q=1$, and the AM–GM inequality answers that. In part (ii) the second round is no longer a fresh start, so we split by what happened in the first round, and the same bound on pq finishes it.

Solution

Part (i)

In any one round the two coins are independent. A decision is made when both show heads (probability p^2) or both show tails (probability q^2), so

$$\mathrm{P}(\text{decision in a given round}) = p^2+q^2, \qquad \mathrm{P}(\text{no decision}) = pq+qp = 2pq.$$

As a check, $p^2+q^2+2pq=(p+q)^2=1$.

The rounds are independent, since fresh tosses are made each time. A decision on the nth round means $n-1$ rounds with no decision followed by one with a decision, so

$$\mathrm{P}(\text{decision on round } n) = (2qp)^{n-1}(q^2+p^2),$$

which is the required result.

For the second result, it is easiest to use the complement. No decision on or before round n means n undecided rounds in a row, with probability $(2pq)^n$. Hence

$$\mathrm{P}(\text{decision on or before round } n) = 1-(2pq)^n.$$

(Summing the geometric series $\sum_{r=1}^{n}(p^2+q^2)(2pq)^{r-1}$ gives the same thing, using $p^2+q^2=1-2pq$.)

Now we need an upper bound on pq that holds for every p. Since $p,q\geq 0$, the AM–GM inequality gives

$$\sqrt{pq}\leq\frac{p+q}{2}=\frac{1}{2},\qquad\text{so}\qquad pq\leq\frac{1}{4},$$

with equality exactly when $p=q=\frac{1}{2}$. (Equivalently, $pq=p(1-p)=\frac{1}{4}-(p-\frac{1}{2})^2$.) Therefore $0\leq 2pq\leq\frac{1}{2}$, so $(2pq)^n\leq\frac{1}{2^n}$ and

$$\mathrm{P}(\text{decision on or before round } n)=1-(2pq)^n\geq 1-\frac{1}{2^n}$$

for every value of p, as required. Fair coins are the worst case.

Part (ii)

Let D_1 be the event of a decision in round 1 and D_2 the event of a decision in round 2. We need $\mathrm{P}(D_1)+\mathrm{P}(D_2)$, since these events are mutually exclusive.

Round 1. All three heads or all three tails: $\mathrm{P}(D_1)=p^3+q^3$.

Round 2. If round 1 did not decide, the coins showed either two heads and one tail or two tails and one head.

- **Two heads, one tail** (probability $3p^2q$, as any of the three people can be the one with the tail). The two who showed heads toss again, and the third turns the tail over to show a head. All tails is now impossible, and all heads happens exactly when both new tosses give heads, with probability p^2. This route contributes $3p^2q\cdot p^2=3p^4q$.
- **Two tails, one head** (probability $3pq^2$). By the same argument, the third coin now shows a tail, and we need both new tosses to give tails, with probability q^2. This route contributes $3pq^2\cdot q^2=3pq^4$.

The round-2 tosses are independent of round 1, which is why we can multiply. So

$$\mathrm{P}(D_2)=3p^4q+3pq^4=3pq\,(p^3+q^3).$$

? **Ask yourself** Is there a neat way to tidy up $p^3+q^3+3pq(p^3+q^3)$?

Factor out p^3+q^3, then use the sum of cubes with $p+q=1$:

$$p^3+q^3=(p+q)(p^2-pq+q^2)=(p+q)^2-3pq=1-3pq.$$

Hence

$$\begin{aligned}\mathrm{P}(\text{decision on or before round 2})&=(p^3+q^3)(1+3pq)\\&=(1-3pq)(1+3pq)\\&=1-9p^2q^2.\end{aligned}$$

From part (i), $0 \le pq \le \frac{1}{4}$, so $p^2q^2 \le \frac{1}{16}$ and

$$\mathrm{P}(\text{decision on or before round 2}) \ge 1 - \frac{9}{16} = \frac{7}{16}$$

whatever the value of p, with equality when $p = \frac{1}{2}$. This completes the proof.

Checks

- With $p = 1$ (double-headed coins) the decision is immediate. In part (i), $1 - (2pq)^n = 1$, and in part (ii), $1 - 9p^2q^2 = 1$, as they should be.
- With $p = \frac{1}{2}$ in part (ii): round 1 decides with probability $\frac{1}{4}$. Otherwise ($\frac{3}{4}$) round 2 decides with probability $\frac{1}{4}$, giving $\frac{1}{4} + \frac{3}{16} = \frac{7}{16}$, so the bound is attained.
- With $p = 0.3$, the formula gives $1 - 9(0.21)^2 = 0.6031$. A computer simulation of 200 000 runs of the three-person procedure gave 0.602.

Common slips

- Forgetting the factor 3 for which person is the odd one out in round 1 of part (ii).
- Treating round 2 of part (ii) like a fresh three-coin round. The turned-over coin is fixed, so only one of "all heads" and "all tails" is possible.
- Proving the bound only at $p = \frac{1}{2}$. You need $pq \le \frac{1}{4}$ for *every* p, and an inequality such as AM–GM gives you that.

Beyond STEP

Part (i) is the classical trick, usually credited to von Neumann, for getting a fair decision from a biased coin. In the two-person procedure, "hotel" and "tea shop" are equally likely only when $p = \frac{1}{2}$, but if one person tosses twice and HT and TH are used as the outcomes, the result is fair for every p. The number of rounds until a decision is a geometric random variable, and questions about how quickly such procedures stop lead to the study of stopping times and Markov chains. Grimmett and Welsh, *Probability: An Introduction*, covers both.

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